

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-FA-8901-03 Sheet 1 of 1
OPL Title	Water Boot Draining (LT-8903)	Discipline	Process / Operations	
Equipment	FA-8901 - REFLUX ACCUMULATOR DRUM	Area / Unit	8900 - HEXANE COLUMN OVERHEAD SYSTEM	
Related Interlock	SEQ-8901 (SIL 1)	P&ID; Ref	TJC-LLD-PID-8901	
Classification:	☐ Basic Knowledge	☐ Improvement	☒ Trouble Case	Aspect: P-Q-C-D-S-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **water boot draining (It-8903)** for FA-8901 (REFLUX ACCUMULATOR DRUM) in the HEXANE COLUMN OVERHEAD SYSTEM. Water accumulating in the boot upsets reflux quality and can slug the reflux pump; regular draining keeps the interface controlled. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Hexane / water** at up to 10 barg / FV / 120 degC. Observe the following before and during the task:

- ☒ Obtain the correct operating permit and brief the shift team before the task.
- ☒ Confirm hazardous fluids are isolated / inerted as required by procedure.
- ☒ Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- ☒ Never defeat a safety interlock or trip without an authorised override permit.
- ☒ Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- ☒ Operating procedure & P&ID;
- ☒ Radio / communication with panel
- ☒ Gas detector (O2 / LEL) as required
- ☒ Correct PPE for the fluid handled
- ☒ Sample containers & PPE for sampling
- ☒ Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Monitor the boot interface level LT-8903	Permit valid & briefed
2	Drain the boot N4 to the oily-water sewer via HV-8904	Parameter within limit
3	Stop draining before hydrocarbon breaks through (watch interface)	Interlock status confirmed
4	Log the drain volume each shift	Change logged on DCS
5	Report a rising water make to process	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Reflux drum level transmitter drifting vs gauge	Transmitter zero drift	Recalibrated LT-8901 against the bridle gauge
Statutory relief valve test	Set pressure drift found	Overhauled PSV-8901, lapped seat, re-set 10 barg, certified
Boot drain valve N4 passing / leaking	Seat eroded by water/solids	Reseated / replaced boot drain valve HV-8904

6. KEY LEARNING POINTS

- ☒ Water accumulating in the boot upsets reflux quality and can slug the reflux pump; regular draining keeps the interface controlled.
- ☒ Respect the interlock SEQ-8901: e.g. LSHH-8901 trips at > 85 % - never defeat it without a permit.
- ☒ Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-8901 before working on FA-8901.
- ☒ Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Monday, 15 June 2026